**th** ∃ решения (1₀), (2)

$\exists\, \varphi(x) \in C(\mathbb{R}^n)$ и огр. $\left(\exists M>0 : |\varphi(x)| \le M \quad \forall x \in \mathbb{R}^n\right)$

тогда реш. з-чи (1₀), (2) ∃ и зад.

$$u(x,t) = \int\limits_{\mathbb{R}^n}\cdots\int \left(\frac{1}{2a\sqrt{\pi t}}\right)^n e^{-\frac{|x-y|^2}{4a^2t}} \cdot \varphi(y)\,dy = \int\limits_{\mathbb{R}^n}\cdots\int K(x-y,t)\,\varphi(y)\,dy$$

ф-ла Пуассона

Δ (продолжение)

$I_2(x,t) \in C(Q_{R,\tau,T})$, $I_2(x,t) \in C^\infty(Q_{R,\tau,T})$

$\exists \frac{\partial}{\partial x_1}(I_2(x,t)) \in C(Q_{R,\tau,T})$

$$\left|\frac{\partial}{\partial x_1}\left(K(x-y,t)\varphi(y)\right)\right| = \left|\frac{\partial}{\partial x_1}\left(\left(\frac{1}{2a\sqrt{\pi t}}\right)^n \cdot e^{-\frac{|x-y|^2}{4a^2t}}\varphi(y)\right)\right| = \left|\left(\frac{1}{2a\sqrt{\pi t}}\right)^n \varphi(y)\, e^{-\frac{|x-y|^2}{4a^2t}} \cdot \left(-\frac{2(x_1-y_1)}{4a^2t}\right)\right| =$$

$$= \underbrace{\left(\frac{1}{2a\sqrt{\pi t}}\right)^n}_{\le \left(\frac{1}{2a\sqrt{\pi\tau}}\right)^n} \underbrace{|\varphi(y)|}_{\le M} \cdot \underbrace{e^{-\frac{|x-y|^2}{4a^2t}}}_{\le e^{-\frac{(|y|-R)^2}{4a^2T}}} \cdot \underbrace{\frac{|x_1-y_1|}{2a^2t}}_{\le \frac{R+|y|}{2a^2\tau}} \le C(R+|y|)\, e^{-\frac{(|y|-R)^2}{4a^2T}}$$

$(x,t) \in Q_{R,\tau,T}$
$|x|<R$, $0<\tau<t<T$
$y : |y|>2R$

$|y|>2R$ ⇒ сход. по пр-ку Вейерштрасса → $I_2 \in C(Q_{R,\tau,T})$

$|x_1-y_1| \le |x_1|+|y_1| \le |x|+|y| \le R+|y|$

Аналогично для более старш. произв. ⇒ $I_2 \in C^\infty(Q_{R,\tau,T})$

$x \in \mathbb{R}^n$, $b>0$

$$\int\limits_{\mathbb{R}^n}\cdots\int e^{-\frac{|y-x|^2}{b}}dy = \iint\limits_{\mathbb{R}^n} e^{-\frac{(y_1-x_1)^2+\ldots+(y_n-x_n)^2}{b}} dy_1\ldots dy_n = \int\limits_{\mathbb{R}^n}\cdots\int e^{-\frac{(y_1-x_1)^2}{b}}\cdot\ldots\cdot e^{-\frac{(y_n-x_n)^2}{b}} dy_1\ldots dy_n =$$

$$= \int\limits_{-\infty}^{+\infty} e^{-\frac{(y_1-x_1)^2}{b}} dy_1 \cdot \ldots \cdot \int\limits_{-\infty}^{+\infty} e^{-\frac{(y_n-x_n)^2}{b}} dy_n = (\sqrt{b}\sqrt{\pi})^n$$

вычислим $\int\limits_{-\infty}^{+\infty} e^{-\left(\frac{y_1-x_1}{\sqrt{b}}\right)^2} dy_1 = \sqrt{b}\int\limits_{-\infty}^{+\infty} e^{-\left(\frac{y_1-x_1}{\sqrt{b}}\right)} d\left(\frac{y_1-x_1}{\sqrt{b}}\right) = \sqrt{b}\int\limits_{-\infty}^{+\infty} e^{-\xi_3^2} d\xi_3 = \sqrt{b}\sqrt{\pi}$

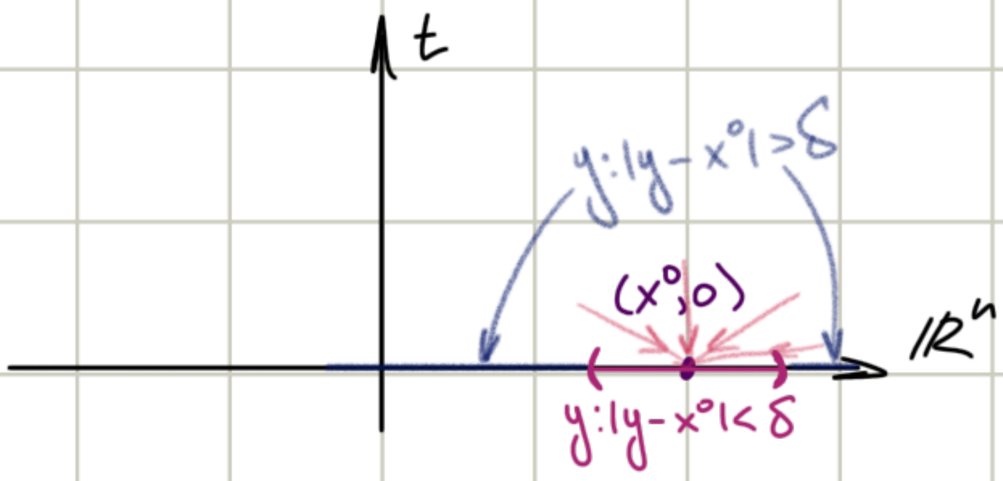
! $x \in \mathbb{R}^n$, $b>0$ ⇒ $\int\limits_{\mathbb{R}^n}\cdots\int e^{-\frac{(y-x)^2}{b}} dy = (\sqrt{\pi}\sqrt{b})^n$

$x \in \mathbb{R}^n$, $t>0$:

$$\int\limits_{\mathbb{R}^n}\cdots\int K(x-y,t)\,dy = \left(\frac{1}{2a\sqrt{\pi t}}\right)^n \int\limits_{\mathbb{R}^n}\cdots\int e^{-\frac{|x-y|^2}{4a^2t}} dy = \left(\frac{1}{2a\sqrt{\pi t}}\right)^n \left(\sqrt{\pi}\sqrt{4a^2t}\right)^n = 1$$

(здесь $4a^2t = b$)

! $x \in \mathbb{R}^n$, $t>0$ ⇒ $\int\limits_{\mathbb{R}^n}\cdots\int K(x-y,t)\,dy = 1$






Тогда $\exists \lim\limits_{\substack{(x,t)\to(x^0,0)\\ t>0}} u(x,t) = \varphi(x^0)$

$\varphi(x) \in C(\mathbb{R}^n) \Rightarrow \varphi(x) \in C(x^0)$: $\forall \varepsilon > 0\ \exists \delta = \delta(\varepsilon) > 0 : \forall \xi : |\xi - x^0| < \delta$

$\Downarrow$

$|\varphi(\xi) - \varphi(x^0)| < \varepsilon$

$$u(x,t) - \varphi(x^0)\cdot 1 = \int\limits_{\mathbb{R}^n} \dots \int k(x-y,t)\varphi(y)\,dy - \int\limits_{\mathbb{R}^n} \dots \int k(x-y,t)\varphi(x^0)\,dy = \int\limits_{\mathbb{R}^n} \dots \int k(x-y,t)(\varphi(y)-\varphi(x^0))\,dy =$$

$$= \int\limits_{|y-x^0|<\delta} \dots \int k(x-y,t)(\varphi(y)-\varphi(x^0))\,dy + \int\limits_{|y-x^0|>\delta} \dots \int k(x-y,t)(\varphi(y)-\varphi(x^0))\,dy$$

$J_1$ $J_2$

$$|J_1| = \left| \int\limits_{|y-x^0|<\delta} \dots \int k(x-y,t)(\varphi(y)-\varphi(x^0))\,dy \right| \le \int\limits_{|y-x^0|<\delta} \dots \int k(x-y,t)\cdot \underbrace{|\varphi(y)-\varphi(x^0)|}_{<\varepsilon}\,dy < \varepsilon \int\limits_{|y-x^0|<\delta} \dots \int k(x-y,t)\,dy \le$$

$$\le \varepsilon \int\limits_{\mathbb{R}^n} \dots \int k(x-y,t)\,dy = \varepsilon \cdot 1 = \varepsilon$$

$|J_1| < \varepsilon$

$u(x,t) - \varphi(x^0) = J_1 + J_2$

$$|J_2| = \left| \int\limits_{|y-x^0|>\delta} \dots \int k(x-y,t)(\varphi(y)-\varphi(x^0))\,dy \right| \le \int\limits_{|y-x^0|>\delta} \dots \int k(x-y,t)\left(\underbrace{|\varphi(y)|}_{\le M} + \underbrace{|\varphi(x^0)|}_{\le M}\right)dy \le 2M\left(\frac{1}{2a\sqrt{\pi t}}\right)^n \cdot$$

$$\cdot \int\limits_{|y-x^0|>\delta} \dots \int e^{-\frac{|x-y|^2}{4a^2t}}\,dy = 2M\left(\frac{1}{2a\sqrt{\pi t}}\right)^n \cdot \int\limits_{|y-x^0|>\delta} \dots \int e^{-\frac{|x-y|^2}{8a^2t}} \underbrace{e^{-\frac{|x-y|^2}{8a^2t}}}_{\le e^{-\frac{\delta^2}{32a^2t}}}\,dy \le 2M\left(\frac{1}{2a\sqrt{\pi t}}\right)^n e^{-\frac{\delta^2}{32a^2t}} \int\limits_{\mathbb{R}^n} \dots \int e^{-\frac{|x-y|^2}{8a^2t}}\,dy \quad (=)$$

$y : |y-x^0| > \delta$

$x \to x^0$

$|x-x^0| < \frac{\delta}{2}$

$$|x-y| = |(x-x^0)-(y-x^0)| \ge ||x-x^0| - |y-x^0|| = |y-x^0| - |x-x^0| \ge \delta - \frac{\delta}{2} = \frac{\delta}{2}$$

$$\Rightarrow |x-y|^2 \ge \left(\frac{\delta}{2}\right)^2 \Rightarrow -\frac{|x-y|^2}{8a^2t} \le -\left(\frac{\delta}{2}\right)^2 \cdot \frac{1}{8a^2t}$$

$$(=)\ 2M\left(\frac{1}{2a\sqrt{\pi t}}\right)^n e^{-\frac{\delta^2}{32a^2t}} \left(\sqrt{\pi}\sqrt{8a^2t}\right)^n = 2M(\sqrt{2})^n e^{-\frac{\delta^2}{32a^2t}} < \varepsilon$$

$t \to 0+$

$|J_2| < \varepsilon$

$$|u(x,t) - \varphi(x^0)| = |J_1 + J_2| \le |J_1| + |J_2| < 2\varepsilon$$

$\Rightarrow \exists \lim\limits_{\substack{(x,t)\to(x^0,0)\\ t>0}} u(x,t) = \varphi(x^0)$ т.е. уд(2) и $C\left(\substack{x\in\mathbb{R}^n\\ t\ge 0}\right)$ ▲.